Statistics for Data Science - 2

Graded assignment Solution- Sept 2024

Week 7

1. Suppose $X_1, X_2, \ldots, X_n$ are n iid random variables with mean μ and variance $\sigma^2 = 16$. Using Chebyshev's inequality, find the minimum value of n such that

$$P(\mid \overline{X} - \mu \mid < 1) > 0.90.$$

Answer: 160

Solution:

$E[\bar{X}] = \mu$, $\text{Var}[\bar{X}] = \dfrac{\sigma^2}{n}$

Note that $\bar{X} = X_1 + X_2 + \ldots + X_n$

By Chebyshev's inequality, we have

$$P\left(\left|\overline{X} - \mu\right| < \delta\right) \geq 1 - \frac{\sigma^2}{n\delta^2}$$

And given

$$\begin{aligned} P(\mid \overline{X} - \mu \mid < 1) &> 0.90 \\ \Rightarrow 1 - \frac{\sigma^2}{n\delta^2} &> 0.90 \\ \Rightarrow 1 - \frac{16}{n} &> 0.90 \\ \Rightarrow 1 - 0.90 &> \frac{16}{n} \\ \Rightarrow 0.10 &> \frac{16}{n} \\ \Rightarrow n &> \frac{16}{0.10} \\ \Rightarrow n &> 160 \end{aligned}$$

Hence, the minimum of n should be 160.

2. Consider a sample 1, 0, 1, 0, 1, 1, 0, 1, 1, 1 from the Bernoulli(0.6) distribution.

   (a) Compute the empirical distribution of the sample.

   A. $p(0) = 0.4, p(1) = 0.6$

   B. $p(0) = 0.3, p(1) = 0.7$

   C. $p(0) = 0.5, p(1) = 0.5$

D. $p(0) = 0.7, p(1) = 0.3$

Answer: B

Solution :
Since, the empirical distribution is the discrete distribution with PMF

$$p(t) = \frac{\#(X_i = t)}{n}$$

$\#(X_i = t)$ denotes the number of times t occurs in the samples.
Therefore,

$$p(0) = \frac{3}{10} = 0.3$$

And

$$p(1) = \frac{7}{10} = 0.7$$

Hence, option (B) is correct.

(b) Compute the sample mean. Enter the answer correct to one decimal place.

Answer: 0.7

Solution:

$$\begin{aligned}\bar{X} &= \frac{X_1 + X_2 + \ldots + X_10}{n} \\ &= \frac{1+0+1+0+1+1+0+1+1+1}{10} \\ &= \frac{7}{10} \\ &= 0.7\end{aligned}$$

3. Let X_1, X_2, X_3 are three independent and identically distributed random variables with mean μ and variance σ^2. Given below are 3 different formulations of sample mean. (Observe that $E[A] = E[B] = E[C]$).

$$A = \frac{X_1 + X_2 + X_3}{3}$$

$$B = 0.1X_1 + 0.3X_2 + 0.6X_3$$

$$C = 0.2X_1 + 0.3X_2 + 0.5X_3$$

Choose the correct option from the following:

(a) Var(A) = Var(B) = Var(C)
(b) Var(A) $\geq$ Var(B) $\geq$ Var(C)
(c) Var(A) $\leq$ Var(B) $\leq$ Var(C)
(d) Var(A) $\leq$ Var(C) $\leq$ Var(B)

Solution:

Let $X_1, X_2, X_3 \sim$ i.i.d. X, where $E[X] = \mu$, $\text{Var}(X) = \sigma^2$

$$\begin{aligned}
\text{Var}(A) &= \text{Var}\left(\frac{X_1 + X_2 + X_3}{3}\right) \\
&= \frac{1}{9}(\text{Var}[X_1] + \text{Var}[X_2] + \text{Var}[X_3]) \\
&= \frac{1}{9}(3\sigma^2) = \frac{\sigma^2}{3}
\end{aligned}$$

$$\begin{aligned}
\text{Var}(B) &= \text{Var}(0.1X_1 + 0.3X_2 + 0.6X_3) \\
&= 0.01\text{Var}[X_1] + 0.09\text{Var}[X_2] + 0.36\text{Var}[X_3] \\
&= 0.46\sigma^2
\end{aligned}$$

$$\begin{aligned}
\text{Var}(C) &= \text{Var}(0.2X_1 + 0.3X_2 + 0.5X_3) \\
&= 0.04\text{Var}[X_1] + 0.09\text{Var}[X_2] + 0.25\text{Var}[X_3] \\
&= 0.38\sigma^2
\end{aligned}$$

Therefore, $\text{Var}(B) \geq \text{Var}(C) \geq \text{Var}(A)$.

4. A random sample of size 25 is collected from a normal population with a mean of 50 and a standard deviation of 5. Find the variance of the sample mean.

Solution:

We know that the variance of the sample mean $\overline{X}$ is given by

$$\begin{aligned}
\text{Var}[\overline{X}] &= \frac{\sigma^2}{n} \\
&= \frac{5^2}{25} \\
&= 1
\end{aligned}$$

5. A fair die is rolled 100 times. Let $\frac{X}{100}$ denote the number of times six is obtained. Find a bound for the probability that $\frac{X}{100}$ differs from $\frac{1}{6}$ by less than 0.1 using the weak law of large numbers.

   (a) at least $\frac{5}{36}$

   (b) at least $\frac{31}{36}$

   (c) at most $\frac{5}{36}$

   (d) at most $\frac{31}{36}$

   Solution:

   X denotes the number of times six is obtained on rolling a fair die 100 times. Let $X_1, X_2, \ldots, X_{100}$ be 100 i.i.d. samples such that

$$X_i = \begin{cases} 1 & \text{if six appears on rolling a fair die} \\ 0 & \text{otherwise} \end{cases}$$

$$E[X_i] = \mu = \frac{1}{6} \quad \text{and} \quad \text{Var}(X_i) = \sigma^2 = \frac{5}{36}$$

   Notice that $X = X_1 + X_2 + X_3 + \cdots + X_{100}$.

   To find: Bound on $P\left(\left|\frac{X}{100} - \frac{1}{6}\right| < 0.1\right)$. By the weak law of large numbers, we have

$$P(|\overline{X} - \mu| < \delta) \geq 1 - \frac{\sigma^2}{n\delta^2}$$

$$\Rightarrow P\left(\left|\frac{X}{100} - \frac{1}{6}\right| < 0.1\right) \geq 1 - \frac{\frac{5}{36}}{100 \times 0.01}$$

$$\Rightarrow P\left(\left|\frac{X}{100} - \frac{1}{6}\right| < 0.1\right) \geq 1 - \frac{5}{36} = \frac{31}{36}$$

6. Let $X_1, X_2, \ldots, X_5$ be i.i.d. samples whose distribution has a mean of 20 and variance of 4. Suppose the sample variance is defined as

$$S^2 = \frac{(X_1 - \overline{X})^2 + \cdots + (X_5 - \overline{X})^2}{5}$$

   where $\overline{X} = \frac{X_1 + X_2 + \cdots + X_5}{5}$. Find the expected value of S^2.

Solution:

$$E[\overline{X}] = \mu = 20 \quad \text{and} \quad \text{Var}[\overline{X}] = \frac{\sigma^2}{n} = \frac{4}{5} = 0.8.$$

$$\begin{aligned}
E[S^2] &= E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2\right] \\
&= \frac{1}{n}E\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] \\
&= \frac{1}{n}E\left[\sum_{i=1}^{n}(X_i^2 + \overline{X}^2 - 2X_i\overline{X})\right] \\
&= \frac{1}{n}E\left[\sum_{i=1}^{n}X_i^2 + n\overline{X}^2 - 2n\overline{X}\overline{X}\right] \\
&= \frac{1}{n}E\left[\sum_{i=1}^{n}X_i^2 + n\overline{X}^2 - n\overline{X}^2\right] \\
&= \frac{1}{n}E\left[\sum_{i=1}^{n}X_i^2\right] \\
&= \frac{1}{n}\left[\sum_{i=1}^{n}E[X_i^2] - nE[\overline{X}^2]\right] \\
&= \frac{1}{n}\left[\sum_{i=1}^{n}(\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right] \\
&= \frac{1}{n}\left[(n\sigma^2 + n\mu^2) - (\sigma^2 + n\mu^2)\right] \\
&= \frac{(n-1)\sigma^2}{n}
\end{aligned}$$

Here, $n = 5$, therefore, $E[S^2] = \frac{4}{5} \times 4 = 3.2$.

7. Suppose $X_i \sim \text{Normal}\left(0, \frac{1}{i^2}\right)$, where $i = 1, 2, \ldots, 9$ and $X_1, X_2, \ldots, X_9$ are independent of each other. Let Y be a random variable defined as $Y = \sum_{i=1}^{9} iX_i$. Find the variance of Y.

   Solution

$$\begin{aligned}
\mathrm{Var}(Y) &= \mathrm{Var}\left(\sum_{i=1}^{9} iX_i\right) \\
&= \mathrm{Var}(X_1 + 2X_2 + 3X_3 + \cdots + 9X_9) \\
&= \mathrm{Var}(X_1) + \mathrm{Var}(2X_2) + \cdots + \mathrm{Var}(9X_9) \\
&= \mathrm{Var}(X_1) + 4\mathrm{Var}(X_2) + \cdots + 81\mathrm{Var}(X_9) \\
&= \frac{1}{1^2} + 4\left(\frac{1}{2^2}\right) + \cdots + 81\left(\frac{1}{9^2}\right) \\
&= 9
\end{aligned}$$

8. A random sample of size 50 is collected from a population P, where $P \sim \text{Uniform}[0, 12]$. Find a lower bound on the probability that the sample mean will be at most 3 units away from the actual mean using the weak law of large numbers.

   Solution:

$$P \sim \text{Uniform}[0, 12]$$
$$E[P] = \mu = \frac{0+12}{2} = 6, \quad \mathrm{Var}(P) = \sigma^2 = \frac{(12-0)^2}{12} = \frac{144}{12} = 12$$

   By weak law of large numbers, we have

$$\begin{aligned}
P(|\overline{X} - \mu| < \delta) &\geq 1 - \frac{\sigma^2}{n\delta^2} \\
P\left(|\overline{X} - \mu| < 3\right) &\geq 1 - \frac{12}{50 \times 9} = \frac{73}{75} \approx 0.9733
\end{aligned}$$

9. Suppose a random sample is used to estimate the proportion of voters in a city. If the sample proportion is roughly 0.45, what sample size is necessary so that the standard deviation of the sample proportion is 0.02?

   Solution

   Let the random variable X represent that the selected candidate is a voter.
   Let X_i be defined as

$$X_i = \begin{cases} 1, & \text{if the selected candidate is a voter} \\ 0, & \text{otherwise} \end{cases}$$

   Define an event A as $A : X = 1$.
   It is given that $P(A) = 0.45$.
   We know that $\mathrm{Var}(S(A)) = \dfrac{P(A)(1 - P(A))}{n}$.

$$\sqrt{\frac{p(1-p)}{n}} = 0.02$$
$$\sqrt{\frac{(0.45)(0.55)}{n}} = 0.02 \implies n = 618.75 \approx 619$$

10. The average life (in years) of an electronic watch follows an exponential distribution with parameter $\frac{1}{2}$. Find the lower bound on the probability that the mean life of a random sample of 50 such watches falls between 1 and 3 years. Enter your answer correct to two decimals.
Hint: Use the weak law of large numbers.

Solution

Let the random variable X represent the life of an electronic watch.
It is given that $X \sim \text{Exp}(1/2)$ and 50 such samples are taken.
$E[X] = \mu = 2, \text{Var}(X) = \sigma^2 = 4$

To find: a lower bound on $P(1 < \overline{X} < 3)$.

By the weak law of large numbers, we have

$$\begin{aligned} P(|\overline{X} - \mu| < \delta) &\geq 1 - \frac{\sigma^2}{n\delta^2} \\ P\left(|\overline{X} - 2| < 1\right) &\geq 1 - \frac{4}{50 \times 1} \\ &= \frac{23}{25} \\ &= 0.92 \end{aligned}$$

11. A university evaluates the final scores of students based on coursework and a final project. The variance of the coursework scores is 15, and the variance of the final project scores is 30. Coursework contributes 70% to the final evaluation, while the project contributes 30%. Assuming the scores of coursework and project are uncorrelated, what is the variance of the final evaluation scores? Enter the answer correct to two decimal places.
Answer : 10.05 ; Range : 10.02 to 10.08

Solution

To compute the variance of the final evaluation scores, we use the formula for the variance of a linear combination of uncorrelated random variables:

$$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y),$$

where X and Y are independent random variables, and a and b are the weights of X and Y, respectively.

In this case:

$$\begin{aligned} a &= 0.7 \quad \text{(weight of coursework)}, \\ b &= 0.3 \quad \text{(weight of the project)}, \\ \text{Var}(X) &= 15 \quad \text{(variance of coursework scores)}, \\ \text{Var}(Y) &= 30 \quad \text{(variance of project scores)}. \end{aligned}$$

Substitute the values into the formula:

$$\text{Var(Final Evaluation)} = (0.7)^2 \cdot 15 + (0.3)^2 \cdot 30.$$

$$\text{Var(Final Evaluation)} = 0.49 \cdot 15 + 0.09 \cdot 30.$$

$$\text{Var(Final Evaluation)} = 7.35 + 2.7 = 10.05.$$

Therefore, the variance of the final evaluation scores is **10.05**.

12. In a large population of students, an unknown proportion p prefers Chips over Cookies. A survey was conducted among 200 students and 120 preferred Chips. Assume the 200 samples are i.i.d. Bernoulli (p).
Based on the given information, answer the following questions:

(a) What is the sample mean? Enter the answer correct to one decimal place.
Answer : 0.6

To determine the sample mean, we use the formula for the sample mean:

$$\bar{X} = \frac{\text{Number of successes}}{\text{Total number of samples}}$$

Given:

$$\begin{aligned} \text{Number of successes} &= 120 \quad \text{(students who preferred Chips)}, \\ \text{Total number of samples} &= 200 \quad \text{(total students surveyed)}. \end{aligned}$$

Substituting the values into the formula:

$$\bar{X} = \frac{120}{200}.$$

Simplify the fraction:

$$\bar{X} = 0.6.$$

Thus, the sample mean is:

$$\boxed{0.6}.$$

(b) What is the variance of the sample mean? Assume p is equal to the sample mean.

A. 0.24
B. 0.36
C. 0.0018
D. 0.0012

Answer : d

The variance of the sample mean is given by the formula:

$$\text{Var}(\bar{X}) = \frac{p(1-p)}{n},$$

where:

$$p = \text{proportion of successes (equal to the sample mean)},$$
$$n = \text{total number of samples}.$$

From the given data:

$$p = 0.6 \quad \text{(calculated sample mean)},$$
$$n = 200 \quad \text{(total number of students surveyed)}.$$

Substituting the values into the formula:

$$\text{Var}(\bar{X}) = \frac{0.6(1-0.6)}{200}.$$

Simplify the expression:

$$\text{Var}(\bar{X}) = \frac{0.6 \cdot 0.4}{200}.$$

$$\text{Var}(\bar{X}) = \frac{0.24}{200}.$$

$$\text{Var}(\bar{X}) = 0.0012.$$

Thus, the variance of the sample mean is:

$$\boxed{0.0012}.$$

(c) If 30 more students join the survey and they all prefer Cookies, what will be the new sample mean and variance of the sample mean?

A. new sample mean = 0.24, new variance = 0.0012
B. new sample mean = 0.6, new variance = 0.0012
C. new sample mean = 0.6522, new variance = 0.0009
D. new sample mean = 0.5217, new variance = 0.00108

Answer : d

The formula for the sample mean is:

$$\bar{X} = \frac{\text{Number of successes}}{\text{Total number of samples}}.$$

From the given data:

$$\text{Original number of successes} = 120,$$
$$\text{Original total number of samples} = 200.$$

If 30 more students join and they all prefer Cookies, the updated values are:

$$\text{New number of successes} = 120 \quad \text{(unchanged, as no additional students prefer Chips)},$$
$$\text{New total number of samples} = 200 + 30 = 230.$$

The new sample mean is:

$$\bar{X} = \frac{120}{230}.$$

Simplify the fraction:

$$\bar{X} = 0.5217 \quad \text{(rounded to 4 decimal places)}.$$

Thus, the new sample mean is:

$$\boxed{0.52174}.$$

The formula for the variance of the sample mean is:

$$\text{Var}(\bar{X}) = \frac{p(1-p)}{n},$$

where:

$$p = \text{new sample mean},$$
$$n = \text{new total number of samples}.$$

Substituting the values:

$$p = 0.52174,$$
$$n = 230.$$

Substitute these into the formula:

$$\text{Var}(\bar{X}) = \frac{0.52174(1-0.52174)}{230}.$$

Simplify the calculation:

$$\text{Var}(\bar{X}) = \frac{0.52174 \cdot 0.47826}{230}.$$

$$\text{Var}(\bar{X}) = \frac{0.249998}{230}.$$

$$\text{Var}(\bar{X}) = 0.00108 \quad \text{(rounded to 5 decimal places)}.$$

Thus, the new variance of the sample mean is:

$$\boxed{0.00108}.$$

13. Let $X_1, X_2, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Uniform}[4, 10]$ and

$$\overline{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}.$$

(a) Which of the following inequalities are true w.r.t Chebyshev's inequality?

A. $P\left(\left|\overline{X} - 7\right| \geq 5\right) \leq \frac{3}{25n}$

B. $P\left(\left|\overline{X} - 7\right| \geq 5\right) \leq \frac{3}{25}$

C. $P\left(\left|\overline{X} - 7\right| \leq 5\right) \geq 1 - \frac{3}{25n}$

D. $P\left(\left|\overline{X} - 7\right| \leq 5\right) \geq \frac{3}{25n}$

Solution:

Since $X_i \sim \text{Uniform}[4, 10]$, we have:

$$\mathbb{E}[X_i] = \frac{4 + 10}{2} = 7, \quad \text{Var}(X_i) = \frac{(10 - 4)^2}{12} = \frac{36}{12} = 3.$$

Then, for the sample mean $\overline{X}$, we have:

$$\mathbb{E}[\overline{X}] = 7, \quad \text{Var}(\overline{X}) = \frac{3}{n}.$$

Using Chebyshev's inequality:

$$P(|\overline{X} - 7| \geq k) \leq \frac{\text{Var}(\overline{X})}{k^2} = \frac{3}{nk^2}.$$

Substitute $k = 5$, we get:

$$P(|\overline{X} - 7| \geq 5) \leq \frac{3}{25n}.$$

Also:

$$\Pr(|\overline{X} - 7| \leq 5) \geq 1 - \frac{3}{25n}.$$

Hence, options (a) and (c) are correct.

Answer: (a), (c)

(b) Find the minimum value of n such that the sample mean lies in $[5, 9]$ with probability more than 0.95 using the Chebyshev inequality.

Solution:

Given that:

$$P(5 < \overline{X} < 9) > 0.95$$

We want:

$$P(|\overline{X} - 7| < 2) > 0.95.$$

By Chebyshev's inequality:

$$P(|\overline{X} - 7| \geq 2) \leq \frac{\text{Var}(\overline{X})}{2^2} = \frac{3}{4n}.$$

Accordingly:

$$\frac{3}{4n} < 0.05 \Rightarrow n > \frac{3}{0.2} = 15.$$

So the minimum integer n is:

$$n = 16.$$

Answer: 16, Range: [15, 17]

14. Suppose $X_1, X_2, X_3, X_4 \sim$ i.i.d. Bernoulli $\left(\frac{2}{3}\right)$. Define three new random variables as:

$$Z_1 = X_1 + X_2 - X_3$$

$$Z_2 = 3X_1 + 2X_2 - 4X_3$$

$$Z_3 = \frac{X_1}{2} + X_2 - \frac{X_3}{2}$$

Choose the correct option from the following:

(a) $\text{Var}(Z_1) < \text{Var}(Z_3) < \text{Var}(Z_2)$

(b) $\text{Var}(Z_2) > \text{Var}(Z_3) > \text{Var}(Z_1)$

(c) $\text{Var}(Z_3) < \text{Var}(Z_1) < \text{Var}(Z_2)$

(d) $\text{Var}(Z_3) = \text{Var}(Z_2) = \text{Var}(Z_1)$

Solution:

Since $X_i \sim$ Bernoulli $\left(\frac{2}{3}\right)$, we have:

$$\mathbb{E}[X_i] = \frac{2}{3}, \quad \text{Var}(X_i) = \frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}$$

As all the random variables are independent and identically distributed,Using the variance property for sum of independent variables

$$Z_1 = X_1 + X_2 - X_3 \Rightarrow \text{Var}(Z_1) = \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) = 3 \cdot \frac{2}{9} = \frac{6}{9} = \frac{2}{3}$$

$$Z_2 = 3X_1 + 2X_2 - 4X_3 \Rightarrow \text{Var}(Z_2) = 9 \cdot \frac{2}{9} + 4 \cdot \frac{2}{9} + 16 \cdot \frac{2}{9} = (2 + \frac{8}{9} + \frac{32}{9}) = \frac{62}{9}$$

$$Z_3 = \frac{X_1}{2} + X_2 - \frac{X_3}{2} \Rightarrow \text{Var}(Z_3) = \frac{1}{4} \cdot \frac{2}{9} + \frac{2}{9} + \frac{1}{4} \cdot \frac{2}{9} = \left(\frac{1}{18} + \frac{2}{9} + \frac{1}{18}\right) = \frac{1}{9} + \frac{4}{18} = \frac{6}{18} = \frac{1}{3}$$

$$\Rightarrow \text{Var}(Z_3) = \frac{1}{3} < \text{Var}(Z_1) = \frac{2}{3} < \text{Var}(Z_2) = \frac{62}{9}$$

Answer: (c)